Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 1

Date: 23-03-2026

PROJECT TITLE

Relational Inventory Control & Stock Tracking System

SDG: SDG No – SDG Title(Example: SDG 4 – Quality Education)

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Courses Handling Faculty:

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INTRODUCTION

1 - Many small and medium businesses still manage inventory manually using notebooks or spreadsheets, leading to errors and inefficiency.

2 Lack of real-time tracking causes stock mismatches, overstocking, and unexpected stockouts.

3 Manual systems increase operational delays and reduce decision-making accuracy.

4 There is a strong need for a centralized and automated inventory management system.

5 This project develops a full-stack web-based Relational Inventory Control & Stock Tracking System.

6 The system allows product management, real-time stock updates, and transaction tracking.

7 It uses a relational database to ensure data consistency and integrity.

8 The solution improves accuracy, efficiency, and overall business performance.

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PROBLEM STATEMENT

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- Many small and medium businesses rely on manual or semi-digital inventory management systems.
- Manual data entry leads to calculation errors and inaccurate stock records.
- There is no real-time synchronization between sales and stock updates.
- Lack of centralized data storage makes tracking and reporting difficult.
- Overstocking and stock shortages occur due to poor monitoring.
- Existing systems do not provide automated alerts for low inventory levels.
- Decision-making becomes slow due to the absence of proper analytics and reports.
- Therefore, there is a need for a reliable, automated, and web-based inventory control system to improve efficiency and accuracy.

OBJECTIVE(S)

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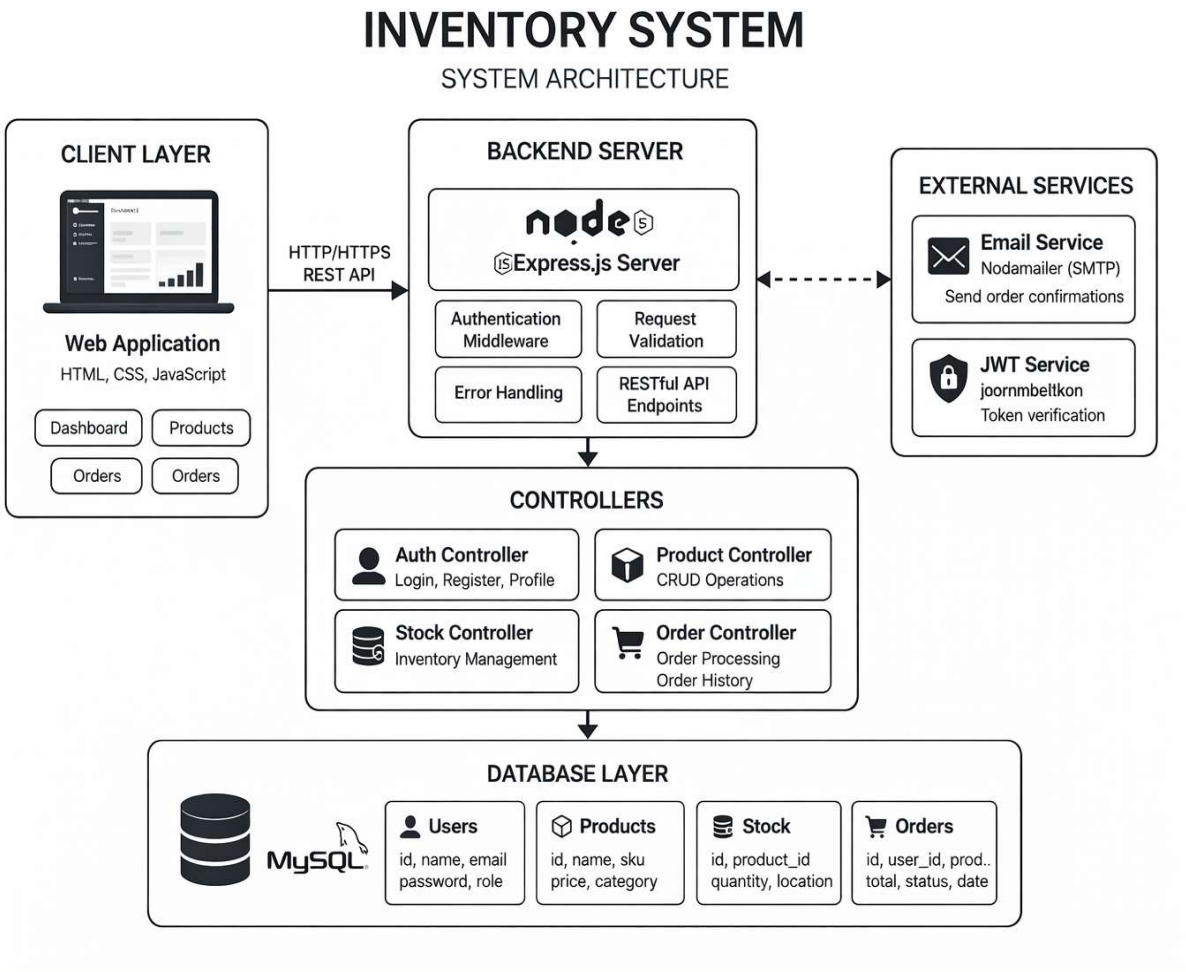
- To design and develop a full-stack web-based Relational Inventory Control & Stock Tracking System using HTML, CSS, JavaScript, backend framework, and MySQL database.

- To implement real-time stock management with automated updates, low-stock alerts, and transaction tracking to improve inventory accuracy and efficiency.

PROPOSED SYSTEM

- The proposed system is a full-stack web application designed to manage inventory and track stock in real time using a relational database.
- It provides secure user authentication for administrators and staff with role-based access control.
- The system allows adding, updating, deleting, and searching products efficiently through a centralized dashboard.
- Stock levels are automatically updated whenever a sales or restocking transaction is recorded.
- Low-stock alerts are generated when product quantity falls below a predefined threshold.
- All product and transaction data are stored in a MySQL relational database to ensure data consistency and integrity.
- The system generates reports and summaries to support faster and data-driven decision-making.
- Compared to manual systems, it reduces human error, improves accuracy, increases operational efficiency, and provides real-time visibility of inventory.

PROPOSED SYSTEM ARCHITECTURE



PROPOSED SYSTEM ARCHITECTURE

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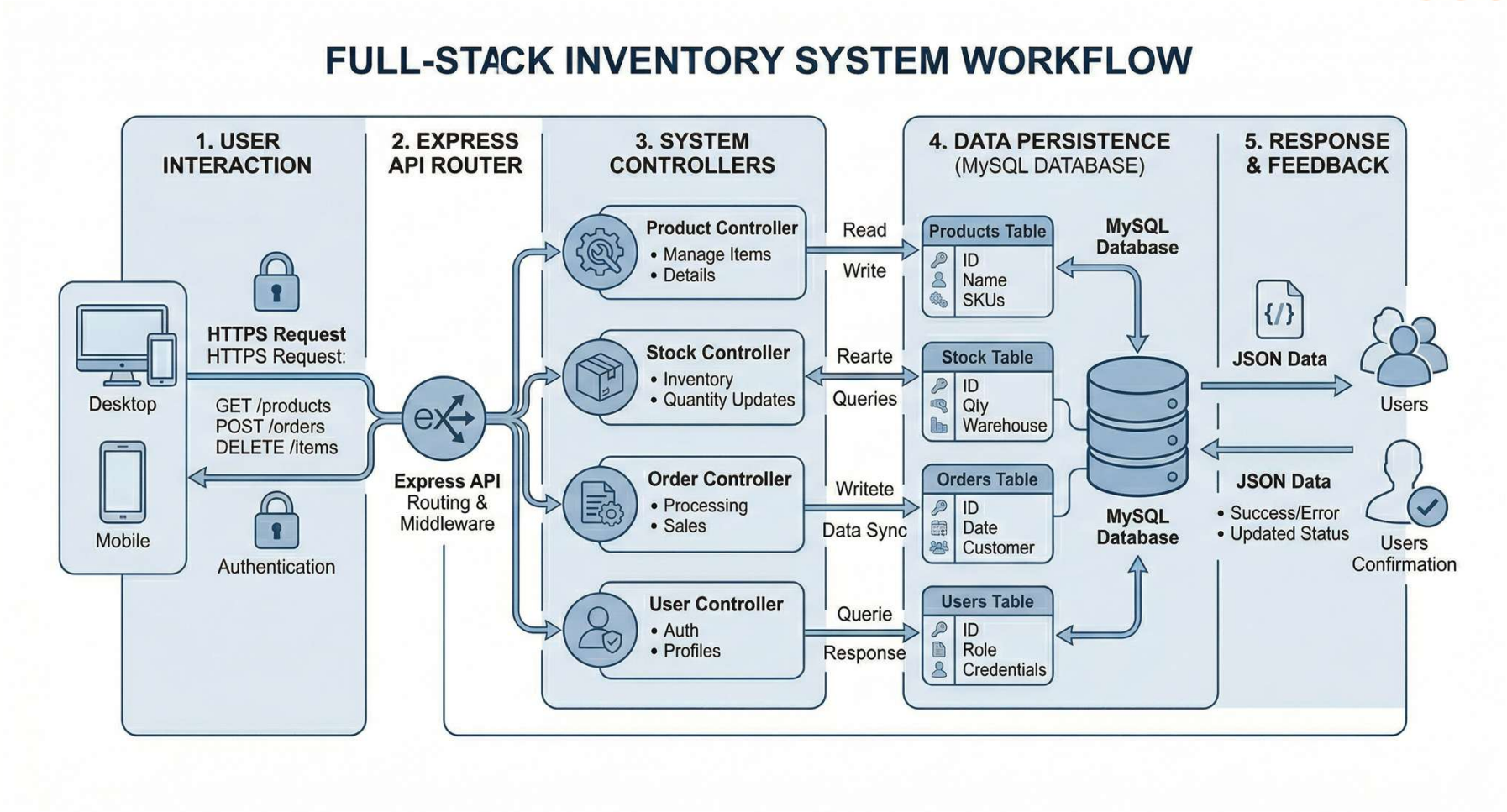
Input
Data given by user through web application
Sent using HTTP / REST API
Examples: Login, Add product, Update stock, Place order
Request goes to backend server
Processing
Done in Node.js / Express server
API routes handle requests
Controllers perform operations
Includes validation and authentication
Auth, Product, Stock, Order controllers
Database
MySQL database used
Stores system data
Tables: Users, Products, Stock, Orders
Server reads and writes data
Output
Server sends response to client
Result shown in dashboard
Examples: Login success, Product added, Order created, Stock updated

WORKFLOW


FULL-STACK INVENTORY SYSTEM WORKFLOW
1. USER INTERACTION
2. EXPRESS API ROUTER
3. SYSTEM CONTROLLERS
4. DATA PERSISTENCE (MySQL DATABASE)
5. RESPONSE & FEEDBACK
Desktop
Mobile
HTTPS Request
HTTPS Request:
GET /products
POST /orders
DELETE /items
Authentication
Express API
Routing & Middleware
Product Controller
• Manage Items
• Details
Stock Controller
• Inventory
• Quantity Updates
Order Controller
• Processing
• Sales
User Controller
• Auth
• Profiles
Read
Write
Rearte
Queries
Writete
Data Sync
Querie
Response
Products Table
ID
Name
SKUs
Stock Table
ID
Qiy
Warehouse
Orders Table
ID
Date
Customer
Users Table
ID
Role
Credentials
MySQL Database
MySQL Database
JSON Data
Users
JSON Data
• Success/Error
• Updated Status
Users Confirmation

TECHNOLOGIES USED

- **Frontend:**
- HTML
- CSS
- JavaScript
- Backend:
- Java
- RESTful APIs
- Database:
- MySQL (Relational Database Management System)
- Tools:
- VS Code
- MySQL Workbench
- Hardware/Software Requirements:
- Minimum 4GB RAM
- Windows / Linux OS
- Web Browser (Chrome/Edge)
- MySQL Server

MODULE DESCRIPTION

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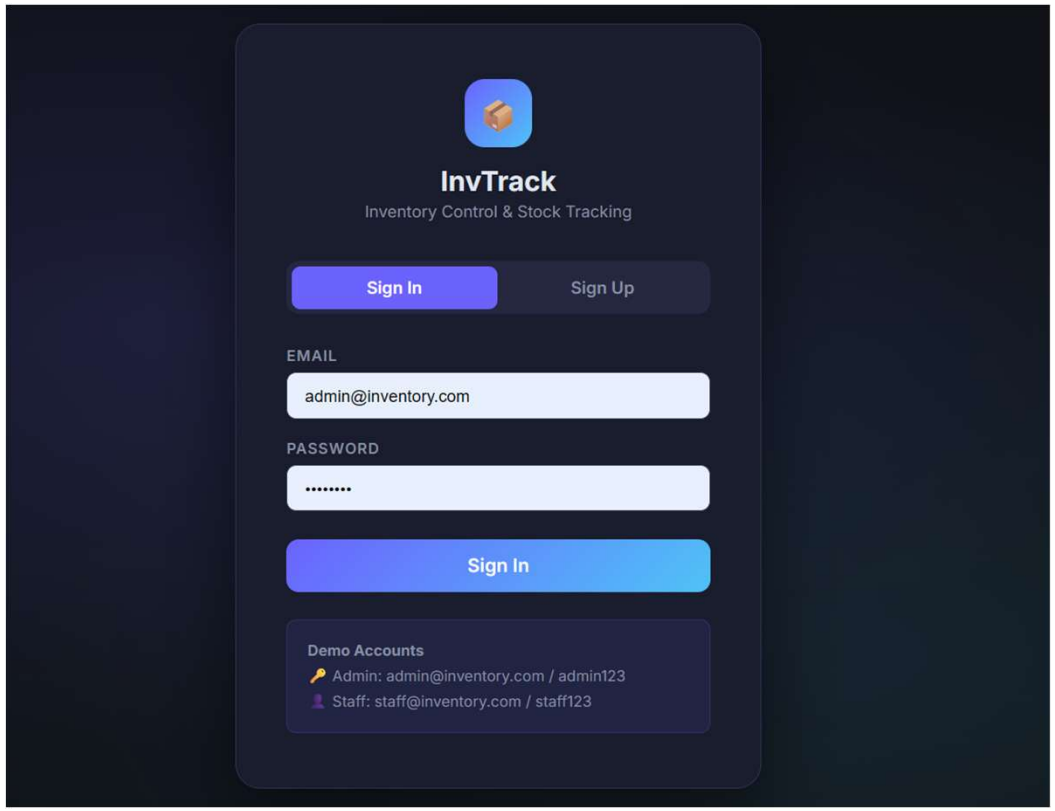
Module 1: User Management
Handles user registration and login using authentication.
Ensures secure access to the system using username and password validation.
Module 2: Product Management
Allows adding, updating, deleting, and viewing product details.
Stores product name, price, category, and quantity information.
Module 3: Stock Management
Manages product stock and quantity updates.
Automatically updates stock when product is added or order is placed.
Module 4: Order Management
Handles order creation and order history.
Updates stock after order processing and stores order details.
Module 5: Database Management
Stores all system data in MySQL database.
Maintains tables for Users, Products, Stock, and Orders.
Module 6: API / Server Processing
Processes requests using Node.js and Express server.
Handles API routes, controllers, and validation.

IMPLEMENTATION SCREENSHOTS



Login Page

This is the login page where the user enters username and password. It provides secure authentication before accessing the system.

IMPLEMENTATION SCREENSHOTS

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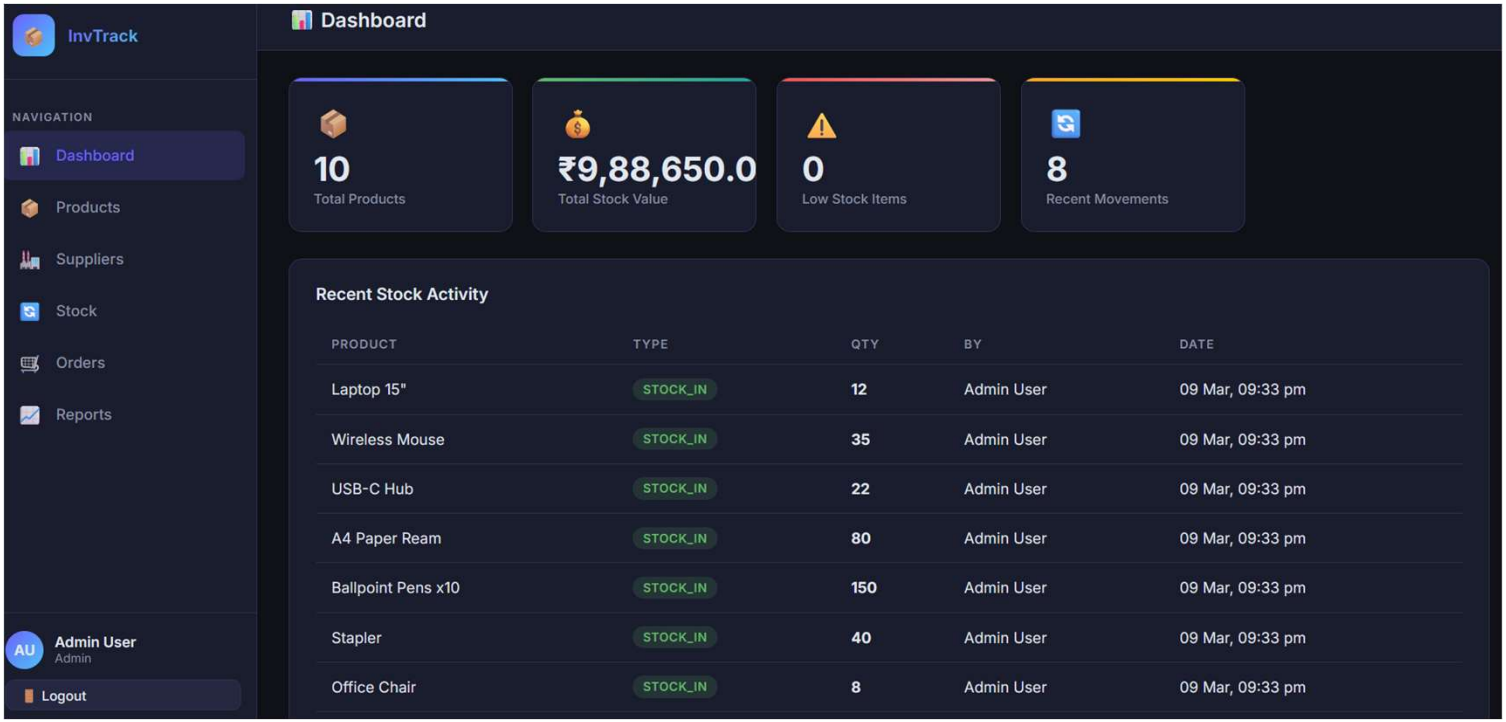

Home Page

This is the home page of our inventory management system.
From here, the user can navigate to login, products, stock, and order modules.

IMPLEMENTATION SCREENSHOTS

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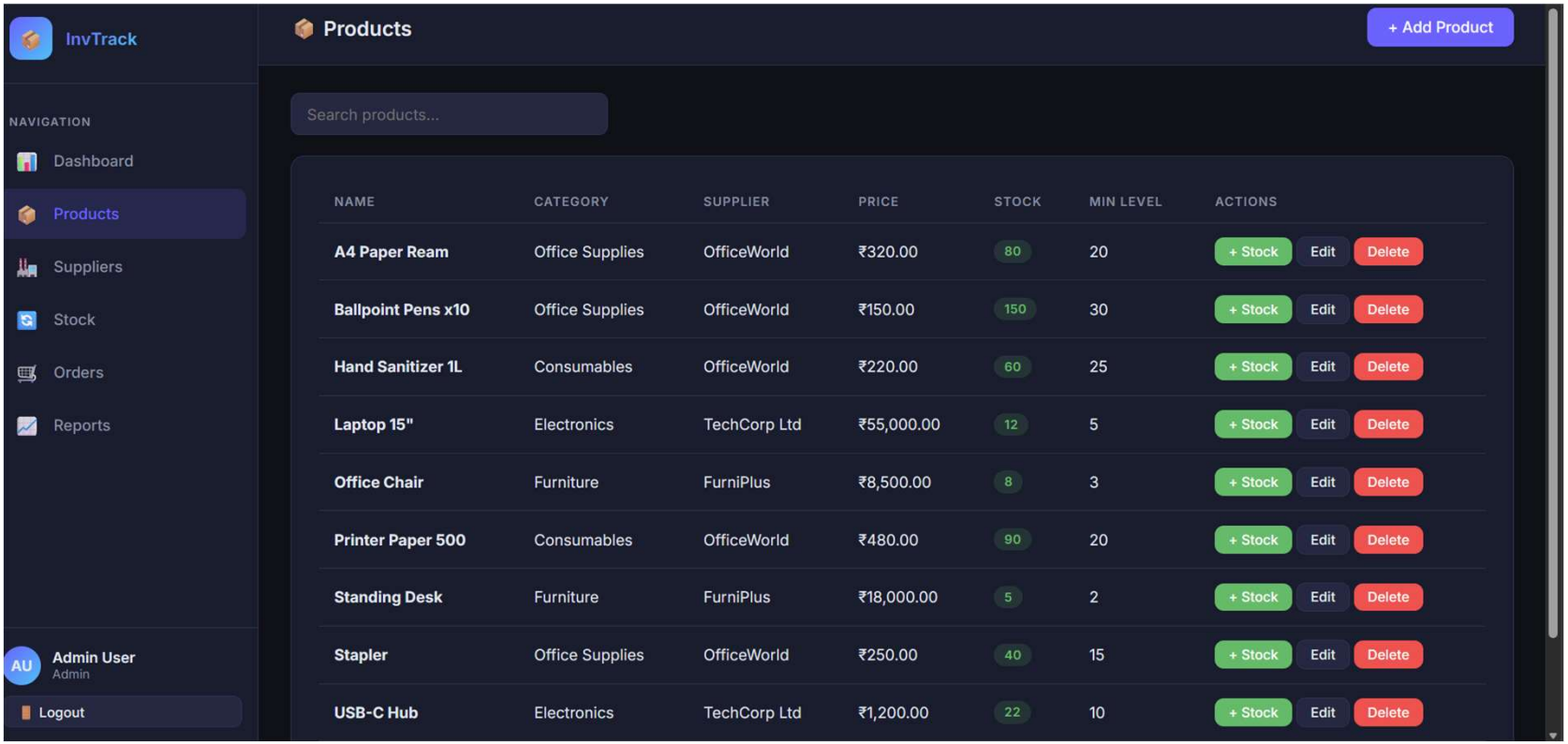

Main Module

This is the main module of the system where we can manage products, stock, and orders. Here we can add, update, delete, and view the details.

IMPLEMENTATION SCREENSHOTS

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Output Page

This is the output page where the result of the operation is displayed.
It shows messages such as product added, stock updated, or order created.

ADVANTAGES & LIMITATIONS

Advantages

- Easy to manage products and stock
- Saves time compared to manual work
- Provides accurate data
- Fast processing using server and database
- Secure login and authentication
- Easy to update and maintain

Limitations

- Requires internet or system to run
- Cannot work if server fails
- Needs basic technical knowledge
- Limited features in current version
- Data loss possible without backup

Thank you